BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024 • Paper Code: [illegible in source scan]

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Mathematics (Science) (Objective Type) - Group - I

Time Allowed: 20 Min. | **Maximum Marks:** 15

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The number of terms in a standard quadratic equation $ax^2 + bx + c = 0$ is:

(A) 1 (B) 3

(C) 2 (D) 4

Q2. Solution set of $5x^2 = 15x$ is:

(A) {0} (B) {5}

(C) {15} (D) {0, 3}

Q3. $\omega =$:

(A) ω^2 (B) ω^{-1}

(C) ω^{-2} (D) ω^{-3}

Q4. If α, β are the roots of $3x^2+5x-2=0$, then $\alpha + \beta$ is :

(A) 5/3 (B) 5/3

(C) -2/3 (D) 2/3

Q5. If a : b = x: y, then invertendo property is :

(A) b/a = y/x (B) b/a = x/y

(C) a/b = y/x (D) a/b = x/y

Q6. If $u \propto v^2$, then :

(A) $u = kv$ (B) $u = k/2$

(C) $u = kv^2$ (D) $u = k^2v$

Q7. $(2x+1)/((x+1)(x-1))$ is:

(A) Equation (B) An identity

(C) Improper fraction (D) Proper fraction

Q8. $\{\emptyset\}$ is ---- set :

(A) Empty (B) Singleton

(C) Universal (D) Domain

Q9. If $A \subseteq B$, then $A \cap B$ is equal to :

(A) $\{\emptyset\}$ (B) $\emptyset$

(C) B (D) A

Q10. The extent of variation between two extreme observations of a data set is measured by :

(A) Variance (B) Mode

(C) Range (D) Average

Q11. $(1/2)\ \text{cosec}\ 45° =$:

(A) $1/(2\sqrt{2})$ (B) $1/\sqrt{2}$

(C) $\sqrt{2}$ (D) $2\sqrt{2}$

Q12. The distance of any point of the circle to its centre is called :

(A) Radius (B) Diameter

(C) Chord (D) Tangent

Q13. A circle has only one ---- :

(A) Chord (B) Radius

(C) Diameter (D) Centre

Q14. Out of two congruent arcs of a circle, if one arc make a central angle of 30°, then the other arc will subtend the central angle of:

(A) 90° (B) 60°

(C) 30° (D) 45°

Q15. The measure of the external angle of a regular octagon is:

(A) $\pi/6$ (B) $\pi/4$

(C) $\pi/3$ (D) $\pi/2$

Mathematics (Science) (Essay Type) (PART - I) - Group - I

Time Allowed: 2.10 hrs. | **Maximum Marks:** 60

Q2. (Marks 12) Write short answers to any SIX (6) questions:

(i) What is meant by radical equation?

(ii) Write the quadratic equation in standard form: $x/(x+1) + (x+1)/x = 6$

(iii) Solve by factorization : $x^2 - x - 20 = 0$

(iv) Find the discriminant of the given quadratic equation: $16x^2 - 24x + 9 = 0$

(v) Evaluate : $\omega^{-13} + \omega^{-17}$

(vi) Write the quadratic equation having following roots : 1,5

(vii) If $R \propto T^2$ and $R = 8$ when $T = 3$, find k

(viii) Define ratio and give one example.

(ix) Find a third proportion to : 6 , 12

Q3. (Marks 12) Write short answers to any SIX (6) questions:

(i) What is meant by an identity?

(ii) Convert into proper fraction : $6x^4/(x^3 + 1)$

(iii) If A and B are two sets, then represent A – B in set builder notation.

(iv) If $Y = Z^+$, $T = O^+$, then find $Y \cup T$

(v) Find number of elements in $Y \times X$, if $X = \{a,b,c\}$ and $Y = \{d, e\}$

(vi) Find $A \times B$ if $A = \{a, b\}$, $B = \{c, d\}$

(vii) Define median.

(viii) Using logarithmic formula, find the geometric mean of 2,4,8

(ix) Find range : 11500,12400,15000,14500, 14800

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) Define coterminal angles.

(ii) Convert - 225° to radian.

(iii) Find r , when $\theta = 75°$, ℓ=52cm.

(iv) Show that : $\sec\theta - \cos\theta = \tan\theta\sin\theta$

(v) Define exterior of a circle.

(vi) Define circumference.

(vii) Define cyclic quadrilateral.

(viii) Draw a circle of radius 4 cm passing through points A and B, 5 cm apart.

(ix) Define escribed circle.

> Note: Note: Attempt any THREE questions in all. But question No. 9 is Compulsory.

Q5. Q.5 (a) Solve the equation $x^2 - 3x - 4 = 0$ by completing square. (b) Prove that : $x^3 + y^3 + z^3 – 3xyz = (x + y + z)(x + wy + w^2z)(x + w^2y + wz)$

(a) Solve the equation $x^2 - 3x - 4 = 0$ by completing square.

(b) Prove that : $x^3 + y^3 + z^3 – 3xyz = (x + y + z)(x + wy + w^2z)(x + w^2y + wz)$

Q6. Q.6 (a) Using theorem of componendo-dividendo , find the value of $(m + 5n)/(m – 5n) + (m + 5p)/(m – 5p)$ if $m = 10np/(n + p)$. (b) Resolve into partial fractions : $(3x + 7)/((x + 3)(x^2 + 4))$

(a) Using theorem of componendo-dividendo , find the value of $(m + 5n)/(m – 5n) + (m + 5p)/(m – 5p)$ if $m = 10np/(n + p)$

(b) Resolve into partial fractions : $(3x + 7)/((x + 3)(x^2 + 4))$

Q7. Q.7 (a) If U = {1, 2, 3, 4, -----,10}, A = {2,4, 6, 8, 10 }, B = { 2,3,5, 7 } then verify $(A \cup B)' = A' \cap B'$. (b) Find mean [Classes: 33-40, 41-50, 51-60, 61-70, 71-75; No. of students: 28, 31, 12, 9, 5]

(a) If U = {1, 2, 3, 4, -----,10}, A = {2,4, 6, 8, 10 }, B = { 2,3,5, 7 } then verify $(A \cup B)' = A' \cap B'$

(b) Find mean. Classes: 33-40, 41-50, 51-60, 61-70, 71-75. No. of students: 28, 31, 12, 9, 5.

Q8. Q.8 (a) If $\cos\theta = -2/3$ and terminal arm of the angle θ is in quadrant III. Find the values of remaining trigonometric functions. (b) Inscribe a circle in a triangle ABC with sides : |AB| = 6cm , |BC| = 3cm , |CA| = 4cm

(a) If $\cos\theta = -2/3$ and terminal arm of the angle θ is in quadrant III. Find the values of remaining trigonometric functions.

(b) Inscribe a circle in a triangle ABC with sides : |AB| = 6cm , |BC| = 3cm , |CA| = 4cm

Q9. Prove that a straight line drawn from the centre of a circle to bisect a chord (which is not a diameter) is perpendicular to the chord. OR. Prove that the measure of a central angle of a minor arc of a circle, is double that of angle subtended by corresponding major arc.

GROUP – II

Mathematics (Science) (Objective Type) - Group - II

Time Allowed: 20 Min. | **Maximum Marks:** 15

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. If $x \propto v^2$, then:

(A) $x = v^2$ (B) $x = kv^2$

(C) $xv^2 = k$ (D) $xv^2 = 1$

Q2. A tangent line intersects the circle at —:

(A) Three points (B) Two points

(C) Single point (D) No point at all

Q3. Sum of the deviations of the variable x from its mean is always :

(A) Zero (B) One

(C) Same (D) Different

Q4. How many common tangents can be drawn for two touching circles :

(A) 3 (B) 2

(C) 4 (D) 5

Q5. solution set of equation $4x^2 - 16 = 0$ is:

(A) $\{\pm 4\}$ (B) $\{4\}$

(C) $\{\pm 2\}$ (D) $\{2\}$

Q6. $(2x+1)/((x+1)(x-1))$ is:

(A) An improper fraction (B) A proper fraction

(C) An equation (D) An identity

Q7. A complete circle is divided into :

(A) 90° (B) 180°

(C) 270° (D) 360°

Q8. $\omega.\omega^2 =$:

(A) ω (B) ω^2

(C) -1 (D) 1

Q9. The length of a chord and the radial segment of a circle are congruent, the central angle made by the chord will be :

(A) 30° (B) 60°

(C) 45° (D) 75°

Q10. Power set of an empty set is :

(A) ∅ (B) {a}

(C) {∅, {a}} (D) {∅}

Q11. If α, β are the roots of $7x^2 - x + 4 = 0$ then αβ is :

(A) -1/7 (B) 4/7

(C) 7/-4 (D) -4/7

Q12. In a ratio x : y , y is called :

(A) Relation (B) Antecedent

(C) Consequent (D) Ratio

Q13. If A = {2,3,5} and B = {3,5,8}, then A ∩ B = :

(A) {3, 5} (B) {5, 8}

(C) {3, 8} (D) {2, 3}

Q14. The solution set of equation $5x^2 = 30x$ is:

(A) {0, 6} (B) {2, 3}

(C) {3, 2} (D) {5, 2}

Q15. $\sec^2\theta$ = ------- :

(A) $1 - \sin^2\theta$ (B) $1 + \tan^2\theta$

(C) $1 + \cos^2\theta$ (D) $1 - \tan^2\theta$

Mathematics (Science) (Essay Type) (PART - I) - Group - II

Time Allowed: 2.10 hrs. | **Maximum Marks:** 60

Q2. (Marks 12) Write short answers to any SIX (6) questions:

(i) Define radical equation.

(ii) Solve by factorization : $3y^2 = y^2 - 5$

(iii) Write in standard form : $1/(x + 4) + 1/(x - 4) = 3$

(iv) Find the discriminant of : $4x^2 - 7x = 2$

(v) Evaluate: $\omega^{37} + \omega^{38} - 5$

(vi) Write the quadratic equation having following roots : -1, -7

(vii) Define direct variation.

(viii) If $w \propto 1/z$ and $w = 5$ when $z = 7$ find k.

(ix) Find a mean proportional between : 20, 45

Q3. (Marks 12) Write short answers to any SIX (6) questions:

(i) Define improper fraction.

(ii) If $x/((x + a)(x - a)) = A/(x + a) + B/(x - a)$, find A and B

(iii) If A = N and B = W, find A - B

(iv) If X = {a,b,c} and Y = {d,e}, find $X \cap Y$ and $X \cup Y$

(v) If L = {a, b, c} and M = {3, 4}, then find a binary relation of $M \times L$.

(vi) Define subset and give an example.

(vii) Define range.

(viii) Write two properties of arithmetic mean.

(ix) Find harmonic mean of data : 12, 5, 8, 4

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) Define an angle.

(ii) Convert $\pi/4$ radian into degree.

(iii) Find 'θ', when $\ell = 2.5$m, r = 4.5 m.

(iv) Simplify each expression to a single trigonometric function $\sin^2 x.\cot^2 x$

(v) Define projection.

(vi) Define collinear points.

(vii) Define cyclic quadrilateral.

(viii) Define perimeter.

(ix) The length of the side of a regular pentagon is 4 cm. What is its perimeter?

> Note: Note: Attempt THREE questions in all. But question No.9 is Compulsory.

Q5. Q.5 (a) Solve the equation $2x^4 = 9x^2 - 4$. (b) If α , β are the roots of the equation $4x^2 - 5x + 6 = 0$ then find the value of $\alpha^2/\beta + \beta^2/\alpha$

(a) Solve the equation $2x^4 = 9x^2 - 4$

(b) If α , β are the roots of the equation $4x^2 - 5x + 6 = 0$ then find the value of $\alpha^2/\beta + \beta^2/\alpha$

Q6. Q.6 (a) Find x in the proportion : $(x – 3)/2 : 5/(x – 1) :: (x – 1)/3 : 4/(x + 4)$. (b) Resolve into partial fractions : $(3x – 11)/((x + 3)(x^2 + 1))$

(a) Find x in the proportion : $(x – 3)/2 : 5/(x – 1) :: (x – 1)/3 : 4/(x + 4)$

(b) Resolve into partial fractions : $(3x – 11)/((x + 3)(x^2 + 1))$

Q7. Q.7 (a) If U = {1,2,3,4, ---, 10}, A = {1, 3, 5, 7, 9}, B = {1, 4, 7, 10} then verify $A – B = A \cap B'$. (b) The marks of six students in Mathematics are as follows. Determine variance. [Student: 1, 2, 3, 4, 5, 6; Marks: 60, 70, 30, 90, 80, 42]

(a) If U = {1,2,3,4, ---, 10}, A = {1, 3, 5, 7, 9}, B = {1, 4, 7, 10} then verify $A – B = A \cap B'$

(b) The marks of six students in Mathematics are as follows. Determine variance. Student: 1, 2, 3, 4, 5, 6. Marks: 60, 70, 30, 90, 80, 42.

Q8. Q.8 (a) If $\operatorname{cosec} \theta = 13/12$ and $\cos \theta > 0$, find the values of $\sin\theta$, $\tan \theta$, $\cot \theta$ and $\sec \theta$. (b) Circumscribe a circle with regard to a right angle triangle with sides 3 cm, 4 cm and 5 cm.

(a) If $\operatorname{cosec} \theta = 13/12$ and $\cos \theta > 0$, find the values of $\sin\theta$, $\tan \theta$, $\cot \theta$ and $\sec \theta$.

(b) Circumscribe a circle with regard to a right angle triangle with sides 3 cm, 4 cm and 5 cm.

Q9. Prove that two chords of a circle which are equidistant from the centre, are congruent. OR Prove that the opposite angles of any quadrilateral inscribed in a circle are supplementary.